PLACEMENT PREPARATION TRACKER

A PROJECT REPORT

Submitted by

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(241401049)

in partial fulfilment of the End Semester Examination for the course

CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY

COMPUTER SCIENCE AND BUSINESS SYSTEM



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled “PLACEMENT PREPARATION TRACKER” is the bonafide work of MADHUMITHA P (241401049), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I, MADHUMITHA P (241401049), hereby declare that the project report titled “PLACEMENT PREPARATION TRACKER”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, dashboard, Deluge scripts, workflows and integration — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai **SIGNATURE OF THE STUDENT**

Date: 03/11/2026

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241401049

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I also acknowledge Zoho Creator Education Programme for providing access to the Zoho Creator platform and learning resources, and I thank my family and friends for their unwavering support.

MADHUMITHA P

ABSTRACT

Campus placement preparation involves many parallel activities: improving aptitude and technical skills, practising programming, building communication ability, maintaining an up-to-date resume, applying to companies and keeping track of interview schedules. When students handle these activities through notebooks, messaging groups and scattered spreadsheets, deadlines are missed, progress cannot be measured and mentors have no reliable view of how ready a student is.

To address these problems, the Placement Preparation Tracker was developed on the Zoho Creator low-code platform. The application holds three forms — Student Details, Preparation Tasks and Company Applications — and links each company application to its student through a lookup field. Deluge scripting automates the key business rules: the student e-mail address is filled automatically when the Student Details form loads, a High priority task cannot be saved without a due date, every new student record is pushed to Zoho Sheet through an authorised Zoho Sheet connection, and the student receives an e-mail when an interview date is entered.

A dashboard presents the average CGPA, the number of students, tasks and company applications, a status summary (Not Started, In Progress, Completed), and charts of priority against completion percentage and year of study against CGPA. Role-based access control separates the Student, Faculty Mentor, Career Counselor, Department Head, Placement Coordinator and System Administrator responsibilities. The application was tested in live mode, promoted from Development to the Stage and Production environments as version 2.0, and secured through permission sets and an authorised connection that keeps credentials out of the scripts. The project shows how a complete, secure and automated placement-preparation system can be delivered quickly with configuration and short Deluge scripts.

Keywords: Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Role-Based Access Control; Zoho Sheet Integration; Placement Preparation.

PROJECT AT A GLANCE

Attribute	Details
Project Title	Placement Preparation Tracker
Application Domain	Education / Campus Placement Preparation Management
Student Name & Register No.	Madhumitha P – 241401049
Zoho Creator Edition	Trial (Zoho Creator Education Programme)
Application Name (Link Name)	Student Placement Tracker (student-placement-tracker)
Published / Live URL	https://creatorapp.zoho.in/madhumitha.p.2024.csbs_ra663/student-placement-tracker/
Demo Credentials (per role)	Admin: madhumitha.p.2024.csbs@rajalakshmi.edu.in UserID: madhumitha.p.2024.csbs@rajalakshmi.edu.in
Number of Forms / Subforms	3 / 0
Number of Reports (by type)	[List: 9] [Kanban: 0] [Calendar: 0] [Pivot/Chart: 0] [Others: 0]
Number of Pages / Dashboards	[1] Dashboard (with Quick Links)
Number of Workflows	[Form: 4] [Schedule: 0] [Approval: 0] [Blueprint: 0] [Batch: 0]
Deluge Functions / Custom Actions	[4]
Integrations	Zoho Sheet Integration (authorised connection)
Roles & Profiles Defined	Student, Faculty Mentor, Career Counselor, Department Head, Placement Coordinator, System Administrator
Environments Used	Development → Stage → Production
Date of Production Deployment	01/11/2026 (Version 2.0)

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CGPA	Cumulative Grade Point Average
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
JSON	JavaScript Object Notation
LCAP	Low-Code Application Platform
LPA	Lakhs Per Annum
OAuth	Open Authorization
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDLC	Software Development Life Cycle
TCO	Total Cost of Ownership
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Placement preparation is one of the most important activities in the final years of an engineering degree. A student has to prepare for aptitude tests, technical interviews, programming tests and communication rounds, keep a resume ready, apply to many companies and remember the dates of assessments and interviews. Mentors such as faculty members, career counselors and placement coordinators in turn need to know how well each student is prepared.

At present most students manage these activities with notebooks, mobile reminders, messaging groups and personal spreadsheets. Information is scattered, there is no common view of progress, and a missed deadline or a forgotten interview date can cost a student an opportunity. To bring all of this into one place, a cloud-based Placement Preparation Tracker was developed using the Zoho Creator low-code application development platform.

1.2 Problem Statement

Manual handling of placement preparation causes scattered records, missed deadlines and poor visibility of progress. Without a centralized system, student details, preparation activities and company applications are stored in different places, so neither students nor mentors can monitor readiness in real time.

- **Scattered Information:** Personal details, resumes, task lists and company information are kept in separate notebooks, files and chats.
- **No Progress Tracking:** Students cannot easily see which preparation tasks are pending, in progress or completed, or which are of high priority.
- **Missed Schedules:** Interview and assessment dates are remembered manually and are often forgotten or noticed late.
- **Lack of Mentor Visibility:** Faculty mentors, counselors and department heads have no central dashboard showing the preparation status of students.
- **Repeated Data Entry:** Student details are re-typed into several lists and spreadsheets, causing errors and duplication.

1.3 Objectives

- To maintain student personal and academic details in a structured Student Details form.

- To manage placement preparation tasks by category, priority, status and completion percentage.
- To track company applications, application status and interview schedules for each student.
- To automate activities using Deluge scripting, including task validation, automatic e-mail filling and interview notifications.
- To provide reports and a dashboard that summarise student, task and company application information.
- To integrate Zoho Creator with Zoho Sheet so that student records are transferred automatically.
- To provide secure access through Role-Based Access Control (RBAC) for six user roles.
- To deploy the application through Development, Stage and Production environments.

1.4 Scope of the Project

The project covers the design, automation and deployment of the Placement Preparation Tracker on Zoho Creator, accessible through web browsers and the Zoho Creator mobile app for students, faculty mentors, career counselors, department heads, placement coordinators and the system administrator.

In-Scope:

- **Forms:** Student Details, Preparation Tasks and Company Applications, with a lookup from Company Applications to Student Details.
- **Workflows:** Auto-fill Student Email, Validate High Priority Tasks, Sync Student Details To Sheet and Notify on Interview Scheduled.
- **Pages & Reports:** Dashboard page with KPI tiles, status summary, two charts and an embedded Preparation Tasks report, plus reports for students, tasks and company applications.
- **Integration & RBAC:** Zoho Sheet connection and role-based permissions for six roles.

Out-of-Scope:

- Online aptitude test engine and automatic evaluation of answers.
- Automatic resume analysis using AI.
- Integration with external company recruitment portals.

1.5 Justification for a Low-Code Approach

Developing the Placement Preparation Tracker on Zoho Creator offers significant advantages over traditional pro-code software engineering. Aligned with Unit I principles, Zoho Creator allows citizen developers to design data models, forms, reports and dashboards visually, which greatly reduces development time. Built-in cloud hosting removes the need to provision servers or maintain databases, and the application can be opened on the Zoho Creator mobile app without a separate mobile code base. Declarative role-based permissions provide secure, role-wise access, and Deluge scripts add business rules wherever configuration alone is not enough. Together these features lower the total cost of ownership, as supported by the low-code development literature.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the current setup, a student keeps personal and academic details in a resume file, writes preparation plans in a notebook or phone note, and collects company information from notice boards, e-mails and messaging groups. Aptitude, technical, programming and communication preparation are tracked mentally or on paper, and application and interview dates are noted wherever convenient. Faculty mentors and counselors learn about a student's progress only by asking, and department heads receive consolidated information, if at all, through manually prepared lists. There is no structured record, no automatic reminder and no shared view of preparation progress.

Figure 2.1: As-Is Process Flow (Manual Placement Preparation)

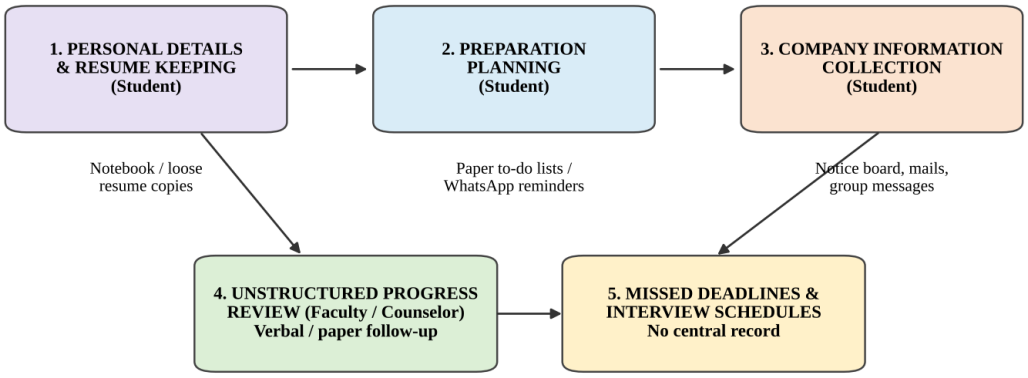


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- **Scattered Records:** Details are spread across notebooks, files and chats, so nothing is available in one place.
- **No Progress Monitoring:** There is no measure of how many preparation tasks are completed or pending.
- **Missed Deadlines:** Due dates and interview schedules depend on memory and are easily forgotten.
- **Repeated Data Entry:** Student information is rewritten for every application and list, causing errors.
- **No Mentor Visibility:** Mentors and department heads cannot review readiness without asking each student.

- **No Notifications:** Updates such as a scheduled interview are not communicated automatically.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Several weeks for design, coding and testing	Days to a couple of weeks using visual builders
Skills required	Full-stack team (front-end, back-end, database, DevOps)	Citizen developer with basic Deluge scripting
Hosting & infrastructure	Developer arranges servers and database	Cloud hosting provided by the platform
Mobile application	Separate native or cross-platform development	Available through the Zoho Creator mobile app
Security & access control	Custom authentication, sessions and middleware	Declarative role and permission configuration
Maintenance & change	Higher effort for dependencies and redeployment	Low effort; changes published from the builder
Customisation limits	Complete freedom over front-end and back-end	Guided by platform boundaries and Deluge limits
Cost	Team salaries and infrastructure costs	Predictable platform subscription

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Gartner Research (2024)	Magic Quadrant for Enterprise Low-Code Application Platforms	Low-code platforms speed up delivery of business applications and reduce reliance on hand-written code.	Supports using Zoho Creator for rapid cloud deployment of the tracker.
2	W. Rosam and O. Kalnichevski	Low-Code Development: A Practical Guide	Explains visual modelling and script-based extension of low-code applications.	Guides the use of forms, reports and Deluge scripts.
3	Zoho Corporation (2025)	Zoho Creator Help Documentation	Lookup fields connect records across forms and preserve relational integrity.	Basis for linking Company Applications to Student Details.
4	Zoho Corporation (2025)	Deluge Script Reference Guide	Deluge validates data, sends e-mail and calls Zoho services from form events.	Foundation of the task validation, e-mail and Zoho Sheet scripts.
5	Zoho Corporation (2025)	Zoho Creator Environments and Governance	Development, Stage and Production environments allow safe testing before publishing.	Used to deploy version 2.0 of the application.

2.5 Emerging Trends in Low-Code Development

Modern low-code platforms combine AI-assisted development, hyper-automation, citizen development and composable architectures. In this project, the Zia Build Agent was used to review the application and correct a field label, while event-driven Deluge workflows automated validation, e-mail notifications and data transfer to Zoho Sheet. Visual drag-and-drop builders allowed the forms, reports and dashboard to be designed quickly, and the modular form structure keeps the application easy to extend.

2.6 Proposed System

The proposed Placement Preparation Tracker centralizes student information, preparation tasks and company applications in one cloud application on Zoho Creator. It removes repeated data entry by linking each company application to its student through a lookup. A validation workflow ensures that High priority tasks always carry a due date, an automatic notification informs the student when an interview date is entered, and each new student record is transferred to Zoho Sheet. A dashboard and reports give students and mentors a live view of preparation progress, replacing notebooks and scattered spreadsheets with structured, automated workflows.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Student	Placement aspirant	Maintains personal details, preparation tasks and company applications.
Faculty Mentor	Teaching staff	Monitors student preparation progress and provides guidance.
Career Counselor	Career guidance staff	Reviews student profiles and tracks career readiness.
Department Head	Head of department	Monitors department-wise performance and placement activity.
Placement Coordinator	Placement cell staff	Manages company details, job applications and placement activities.
System Administrator	System owner	Manages application settings, users, permissions and records.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	Store student name, roll number, e-mail, contact number, department, year, CGPA, skills and resume.	Student, Admin	High
FR-02	Create preparation tasks with category, dates, priority, status and completion percentage.	Student, Admin	High
FR-03	Block saving of a High priority task that has no due date.	Student, Admin	High
FR-04	Record company applications with job role, location, package, status and interview date.	Student, Coordinator	High
FR-05	Send an e-mail to the student when an interview date is entered.	Student, Coordinator	Medium
FR-06	Transfer each submitted student record to Zoho Sheet automatically.	Student, Admin	Medium
FR-07	Display a dashboard with KPI tiles, status summary and charts.	All roles	Medium
FR-08	Provide role-based access to forms, reports and the dashboard.	Admin	High

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports and dashboard load within 2 seconds.
NFR-02	Security	Role-based access control for all user roles.
NFR-03	Usability	Accessible on web and the Zoho Creator mobile app.
NFR-04	Availability	Cloud hosted with platform-level availability.
NFR-05	Maintainability	Easy low-code schema and script updates.
NFR-06	Integrity	Mandatory and format validation on form data entry.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (Trial / Education Programme)
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
Other Zoho services	Zoho Sheet (integration) / Zoho Mail (e-mail notifications)
AI assistance	Zia Build Agent in the Zoho Creator builder
Mobile	Zoho Creator mobile app (Android / iOS) [not tested yet]
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

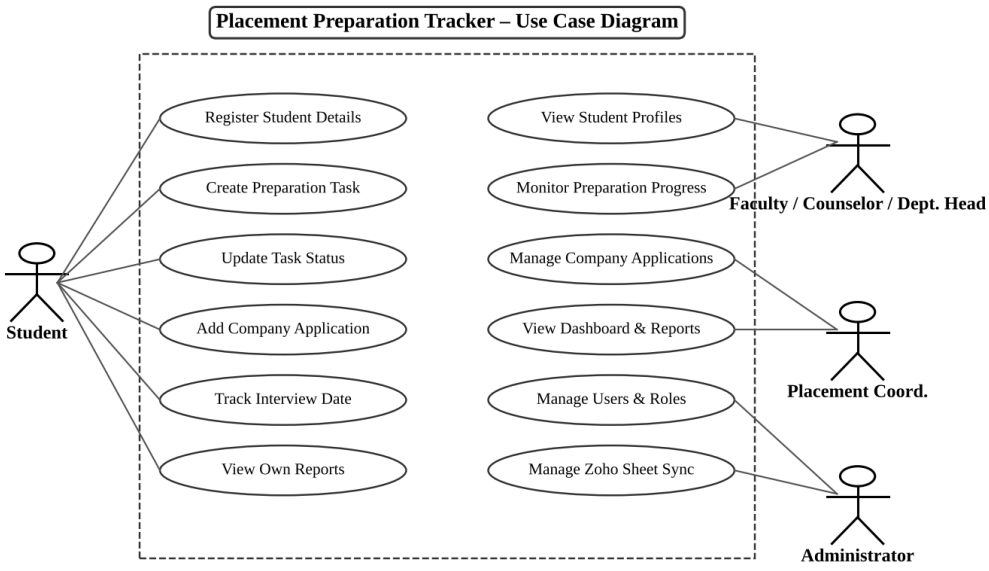


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Create High Priority Preparation Task

Use Case ID	UC-01
Use Case Name	Create High Priority Preparation Task
Actor(s)	Student
Pre-condition	Student is logged in and has access to the Preparation Tasks form.
Main Flow	1. Open Preparation Tasks form. 2. Enter task name, category, dates and priority. 3. Submit the task.
Alternate Flow	If priority is High and the due date is empty, an alert is shown and submission is cancelled.
Post-condition	Task record is created and the dashboard counts are updated.
Related FR	FR-02, FR-03

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

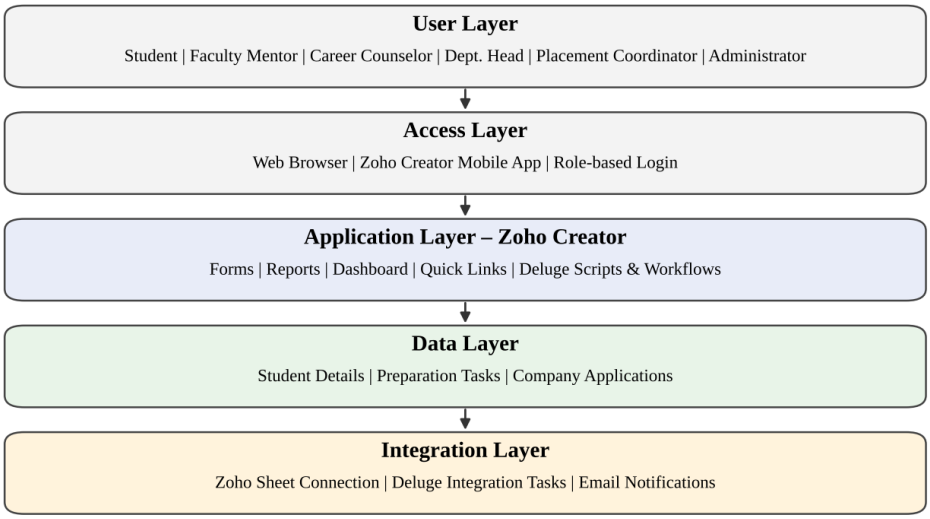


Figure 4.1: System Architecture of the Application

The solution operates across five integrated layers: users in the User Layer (students, mentors, counselors, department heads, coordinators and administrator) reach the Access Layer (web browser and Zoho Creator mobile app) and log in with role-based permissions. The Application Layer executes the operational logic using forms, reports, the dashboard and Deluge workflows. All records are saved in the Data Layer in the Student Details, Preparation Tasks and Company Applications forms, while the Integration Layer manages the Zoho Sheet connection, integration tasks and e-mail notifications.

4.2 Data Model

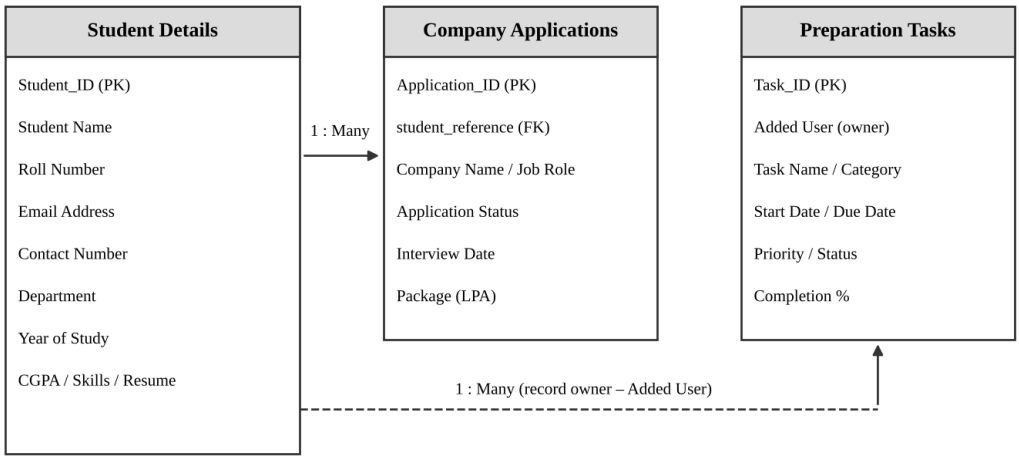


Figure 4.2: Entity–Relationship Diagram

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Student Details	Student Name, Roll Number, Email Address, Contact Number, Department, Year of Study, CGPA, Skills, Resume	Company Applications	One-to-Many	Lookup field “student_reference” in Company Applications
Preparation Tasks	Task Name, Task Category, Task Description, Start Date, Due Date, Priority, Status, Completion Percentage, Remarks	Student (record owner)	One-to-Many	Record owner (Added User) and role-based data sharing
Company Applications	Company Name, Job Role, Job Location, Application Date, Application Type, Job Description, Required Skills, Package (LPA), Application Status, Interview Date, Notes	Student Details	Many-to-One	Lookup field “student_reference”

4.3 Form Design

Table 4.2: Field Specification – Student Details Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Student Name	Student_Name	Single Line	Yes	–	Full name of the student
Roll Number	Roll_Number	Single Line	Yes	Unique roll number	e.g. 241401049
Email Address	email_address	Email	Yes	Auto-filled on load	Used by notifications
Contact Number	Contact_Number	Phone	No	Country code +91	Mobile number
Department	Department	Drop Down	Yes	–	e.g. CSBS
Year of Study	Year_of_Study	Drop Down	Yes	–	1st to 4th Year
CGPA	CGPA	Decimal	Yes	Two decimal places	Used in dashboard average
Skills	Skills	Multi Line	No	–	Technical skills
Resume	Resume	File Upload	No	–	Resume document

Table 4.3: Field Specification – Preparation Tasks Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Task Name	Task_Name	Single Line	Yes	–	Name of the task
Task Category	Task_Category	Drop Down	Yes	–	Aptitude, Technical Skills, Programming, Communication Skills, Interview Preparation, Resume Building
Task Description	Task_Description	Multi Line	No	–	Details of the task
Start Date	Start_Date	Date	Yes	–	dd-MMM-yyyy
Due Date	due_date	Date	Conditional	Required if priority is High	Checked by workflow
Priority	priority	Drop Down	Yes	–	High / Medium / Low
Status	Status	Drop Down	Yes	Default: Not Started	Not Started / In Progress / Completed
Completion Percentage	Completion_Percentage	Number	No	0–100	Progress of the task
Remarks	Remarks	Multi Line	No	–	Student notes

Table 4.4: Field Specification – Company Applications Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Student Reference	student_reference	Lookup	Yes	Lookup to Student Details	Links the application to a student
Company Name	Company_Name	Single Line	Yes	–	Name of the company
Job Role	Job_Role	Single Line	Yes	–	Role applied for
Job Location	Job_Location	Single Line	No	–	City / location
Application Date	Application_Date	Date	Yes	–	Date of applying
Application Type	Application_Type	Drop Down	No	–	On-campus / Off-campus
Job Description	Job_Description	Multi Line	No	–	Role description
Required Skills	Required_Skills	Multi Line	No	–	Skills asked by company
Package (LPA)	Package_LPA	Decimal	No	–	Offered package
Application Status	Application_Status	Drop Down	Yes	Default: Applied	Applied, Assessment, Interview Scheduled, Selected, Rejected
Interview Date	interview_date	Date	No	Triggers notification	User input starts workflow
Notes	Notes	Multi Line	No	–	Additional remarks

4.4 Process and Workflow Design

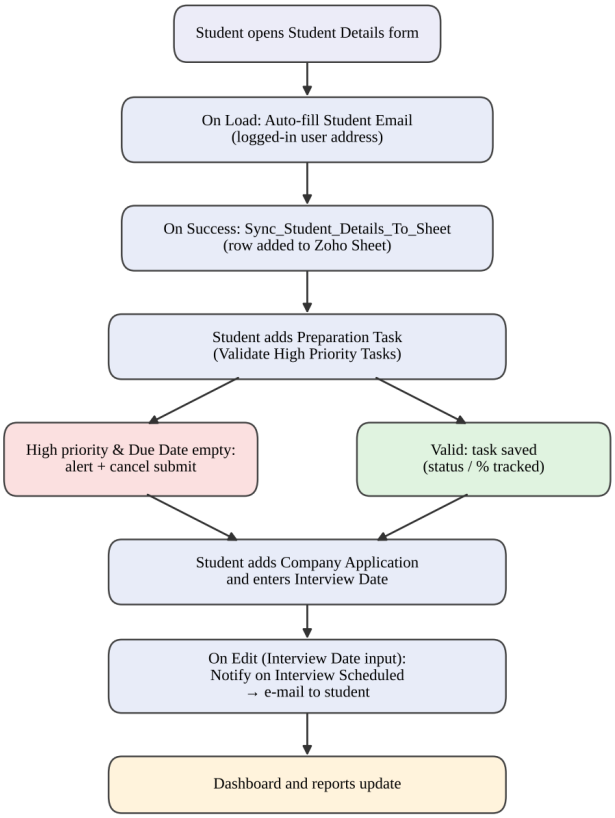


Figure 4.3: Workflow / Process Flow for Placement Preparation Tracking

4.5 User Interface and Navigation Design

The left menu is organised into Dashboard, Preparation Tasks (form, Preparation Tasks Report, Task Category, Tasks by Status, All Preparation Tasks and High Priority Tasks), Student Details (form and report), Company Applications and App Preferences. The dashboard opens first and shows the KPI tiles, the status summary and Quick Links for adding new records.

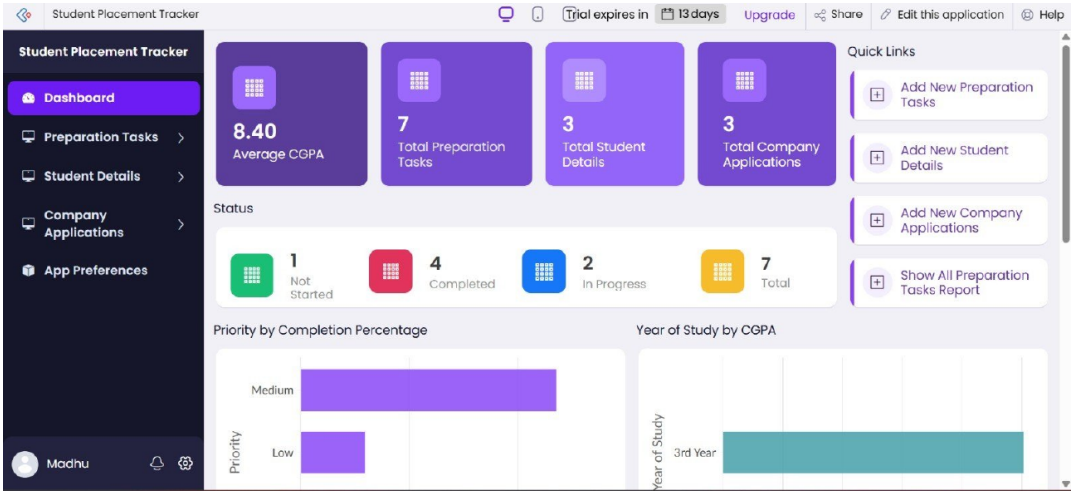


Figure 4.4: Navigation Structure and Wireframe of dashboard

4.6 Access Control Design

Table 4.5: Role–Permission Matrix

Component	Student	Faculty Mentor	Career Counselor	Dept. Head	Placement Coord.	System Admin
Form: Student Details	Acc E V	Acc V	Acc V	Acc V	Acc V	Acc E V D
Form: Preparation Tasks	Acc E V	Acc V	Acc V	Acc V	Acc V	Acc E V D
Form: Company Applications	Acc E V	Acc V	Acc V	Acc V	Acc E V	Acc E V D
Reports: Students and Tasks	V	V	V	V	V	V
Reports: Company Applications	V	V	V	V	V	V
Page: Dashboard	Acc	Acc	Acc	Acc	Acc	Acc
App Settings and Users	–	–	–	–	–	Acc E V D

Acc = Access, V = View, E = Edit, D = Delete, – = no access.

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application “Placement Preparation Tracker” (link name student-placement-tracker) was created in a Zoho Creator workspace on the trial edition by starting from scratch. The workspace date and time settings are IST, dd-MMM-yyyy and 24-hour. The left menu was arranged in the App Menu Builder into the sections Dashboard, Preparation Tasks (form, Preparation Tasks Report, Task Category, Tasks by Status, All Preparation Tasks and High Priority Tasks), Student Details (form and report), Company Applications and App Preferences. Figure 5.1 shows the application home with its navigation sections.

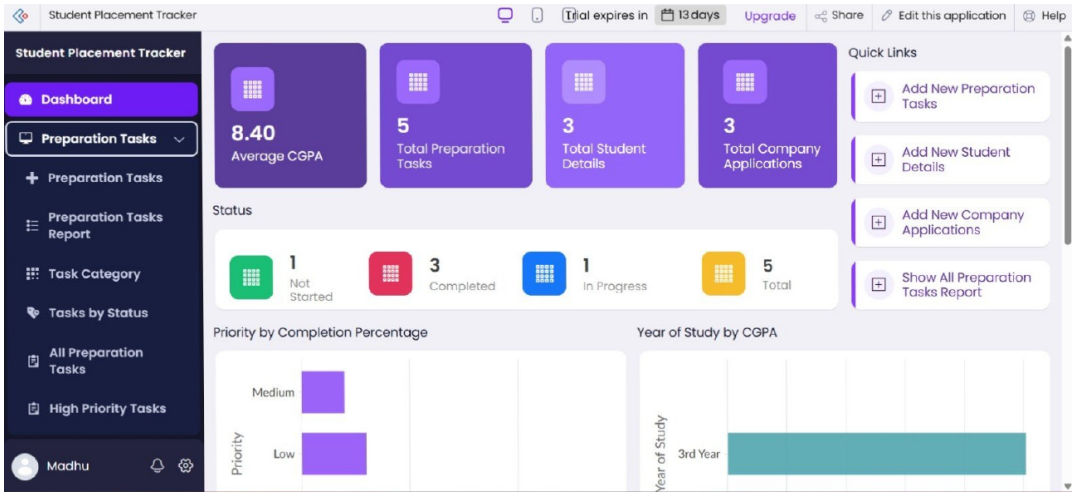


Figure 5.1: Application Home and Navigation Sections

5.2 Forms and Data Capture

Three forms were built (Tables 4.2 to 4.4). Student Details stores the student name, roll number, e-mail address, contact number, department, year of study, CGPA, skills and resume. Preparation Tasks stores the task name, category, description, start and due dates, priority, status, completion percentage and remarks. Company Applications stores the company name, job role, location, application date and type, job description, required skills, package, application status, interview date and notes. Field types used include Single Line, Multi Line, Email, Phone, Number, Decimal, Date, Drop Down, Lookup and File Upload. The Company Applications form uses one lookup field (student_reference) to Student Details. The e-mail address on the Student Details form is filled automatically when the form loads. Figure 5.2 shows the Student Details form in live mode.

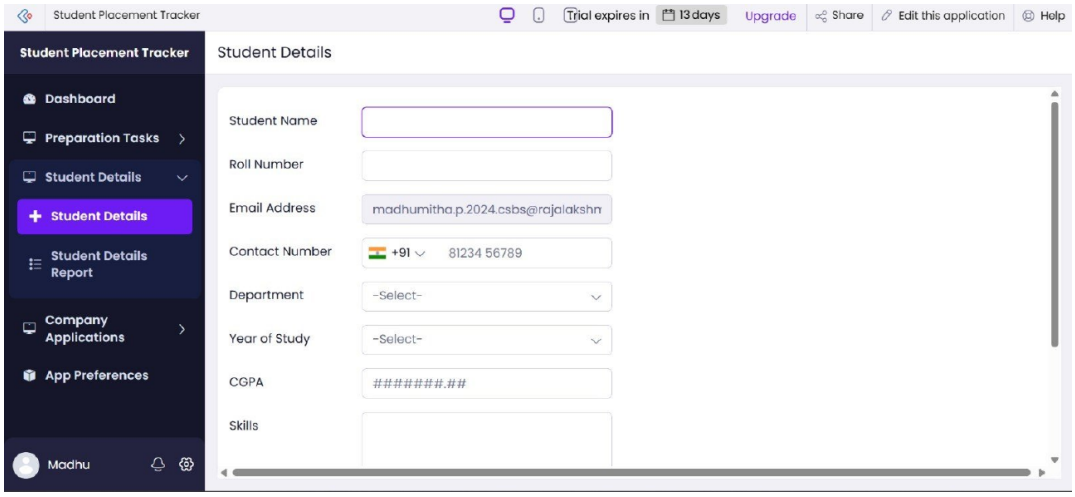


Figure 5.2: Student Details Form in Live Mode

5.3 Reports

Reports display and organise the records stored in the application. Priority values in the Preparation Tasks Report are colour-coded (High, Medium and Low) so that urgent tasks are noticed immediately.

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Roles
Student Details Report	List	Student Details	None (all records)	All roles
Preparation Tasks Report	List	Preparation Tasks	None; colour-coded priority	All roles
All Preparation Tasks	List	Preparation Tasks	None (all records)	All roles
Task Category	List	Preparation Tasks	Grouped by Task Category	All roles
Tasks by Status	List	Preparation Tasks	Grouped by Status	All roles
High Priority Tasks	List	Preparation Tasks	Filter: Priority is High	All roles
All Company Applications	List	Company Applications	None (all records)	All roles
Applications by Status	List	Company Applications	Grouped by Application Status	Coordinator, Mentors
Selected Companies	List	Company Applications	Filter: Status is Selected	Coordinator, Mentors

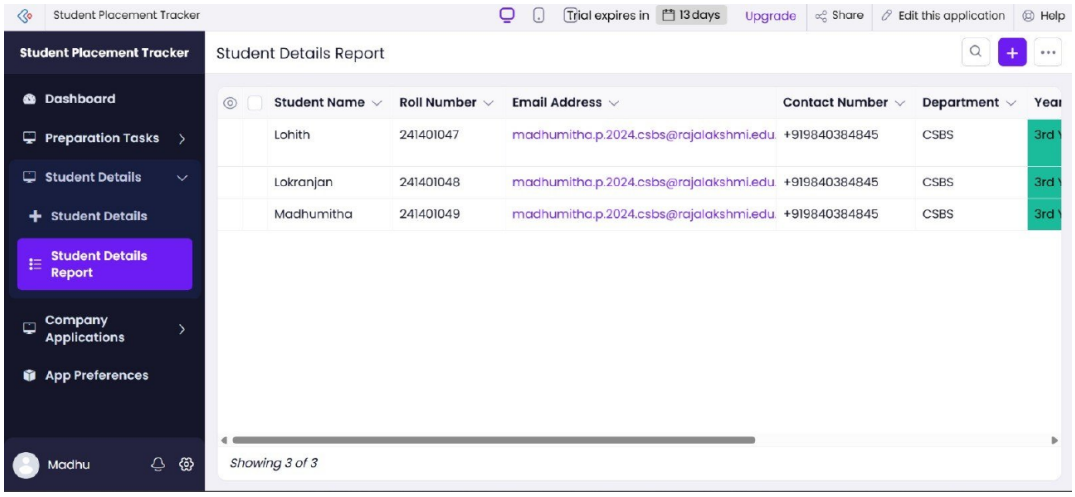


Figure 5.3: Student Details Report

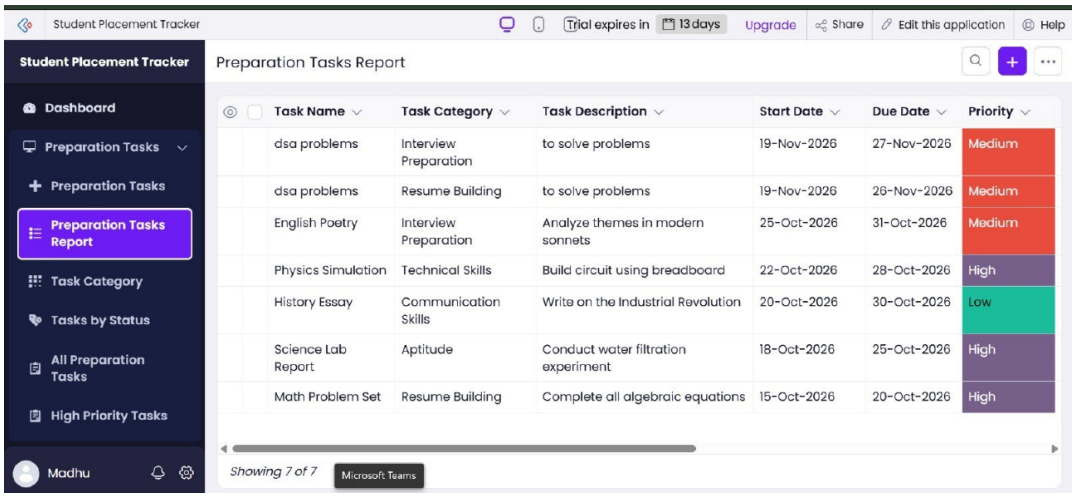


Figure 5.4: Preparation Tasks Report

5.4 Pages and Dashboards

One dashboard page was built with the UI Builder for all roles. It has four KPI tiles (Average CGPA, Total Preparation Tasks, Total Student Details and Total Company Applications), a status summary (Not Started, Completed, In Progress and Total), a bar chart of priority by completion percentage, a bar chart of year of study by CGPA and an embedded Preparation Tasks Report. A Quick Links panel provides buttons to add new preparation tasks, student details and company applications and to open the full Preparation Tasks Report.

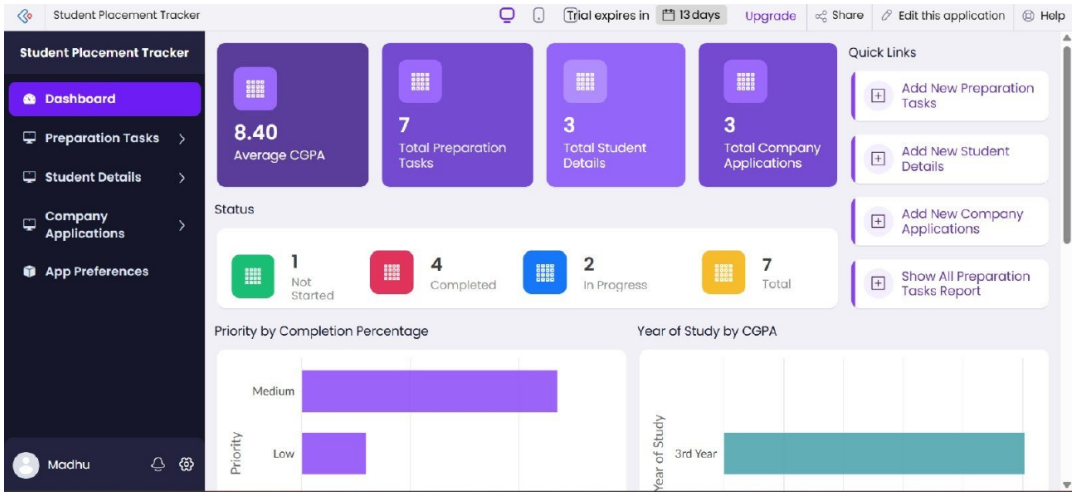


Figure 5.5: Dashboard with KPI Tiles and Status Summary

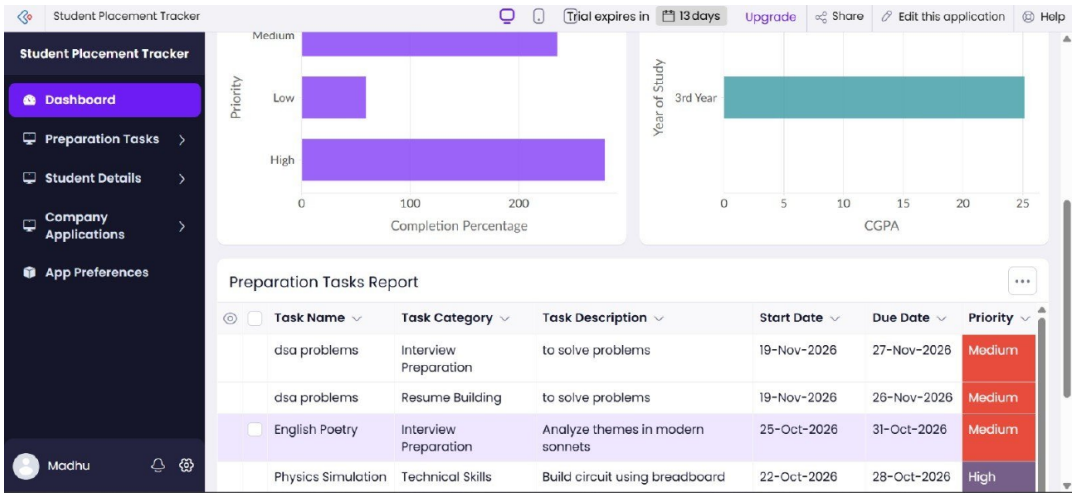


Figure 5.6: Dashboard Charts and Embedded Preparation Tasks Report

5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Student Details → Load of the form	Fill the student e-mail address automatically	input assignment, zoho.loginuserid
DS-02	Preparation Tasks → Created or Edited → Validation on submission	Block a High priority task without a due date	if, isNull(), alert, cancel submit
DS-03	Student Details → Created → Successful form submission	Send the student record to Zoho Sheet	Map, put, zoho.sheet.createRecords, info
DS-04	Company Applications → Edited → User input of Interview Date	E-mail the student when an interview is scheduled	fetch by criteria, if, isNull(), sendmail

DS-01 runs when the Student Details form loads and places the e-mail address of the logged-in user in the e-mail field, so the student does not type it again. DS-02 checks the priority and due date when a task is saved; if the priority is High and the due date is empty, an alert is displayed and cancel submit stops the record from being saved.

The result observed is that a High priority task with a due date is saved normally, while one without a due date is rejected with the alert.

DS-03 runs after a student record is saved. It collects the student name, roll number and e-mail address in a Map and sends them to Zoho Sheet through the authorised Zoho Sheet connection, so that the record appears as a new row in the spreadsheet. DS-04 runs when an interview date is entered in a company application. It fetches the student record using the student_reference lookup, checks that an e-mail address is available and sends a notification mail to the student.

Code 5.1: Auto-fill script – DS-01

```
// DS-01 | Form: Student Details | Event: Created > Load of the form
// Purpose: Fill the e-mail address of the logged-in student
input.email_address = zoho.loginuserid;
```

Code 5.2: Validation script – DS-02

```
// DS-02 | Form: Preparation Tasks | Event: Created or Edited > Validations on form
    submission
// Purpose: Do not allow a High priority task without a due date
// Check if the task priority is High and due_date is missing
if(input.priority == "High" && input.due_date.isNull())
{
    alert "High priority tasks must have a due date. Please enter a due date before
        submitting.";
    cancel submit;
}
```

Code 5.3: Zoho Sheet synchronisation script – DS-03

```
// DS-03 | Form: Student Details | Event: On Create > On Success
// Purpose: Send the submitted student record to Zoho Sheet
rowData = Map();
rowData.put("Student Name",input.Student_Name);
rowData.put("Roll Number",input.Roll_Number);
rowData.put("Email Address",input.email_address);
queryData = Map();
response = zoho.sheet.createRecords("<workbook_resource_id>","Student
    Details",rowData,queryData,"zoho_sheet_connection");
info response;
```

Code 5.4: Interview notification script – DS-04

```
// DS-04 | Form: Company Applications | Event: Edited > User input of Interview Date
// Purpose: E-mail the student when an interview date is entered
// Fetch the student record using the student_reference lookup field
studentRecord = student_details[ID == input.student_reference];
// Check if student email address is available
if(!studentRecord.email_address.isNull())
{
    // Send notification email to the student
    sendmail
    [
        from :zoho.loginuserid
        to :studentRecord.email_address
        subject :"Interview Scheduled - " + input.Company_Name
        message :"Hello " + studentRecord.Student_Name + ", your interview with " +
            input.Company_Name + " is scheduled on " + input.interview_date + "."
    ]
}
```

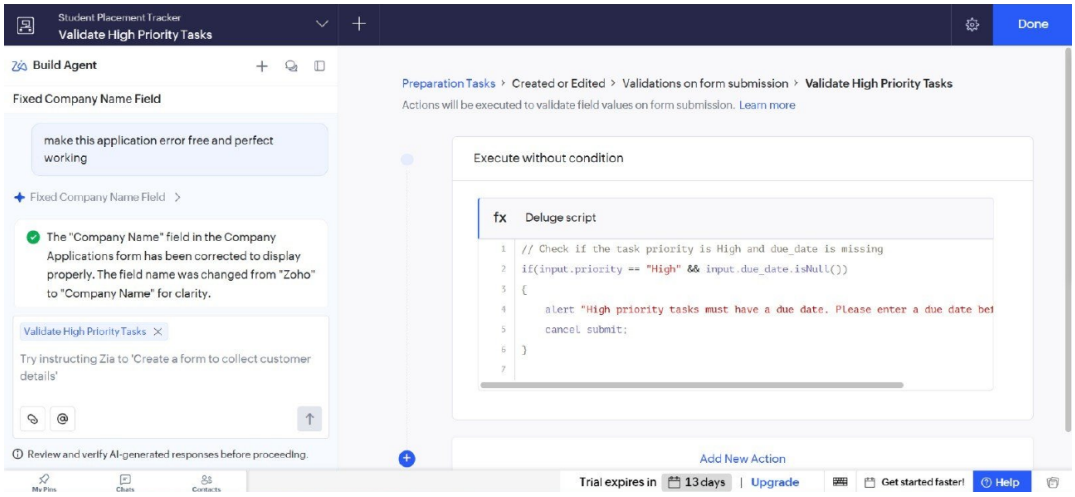


Figure 5.7: Deluge Script of the Validate High Priority Tasks Workflow

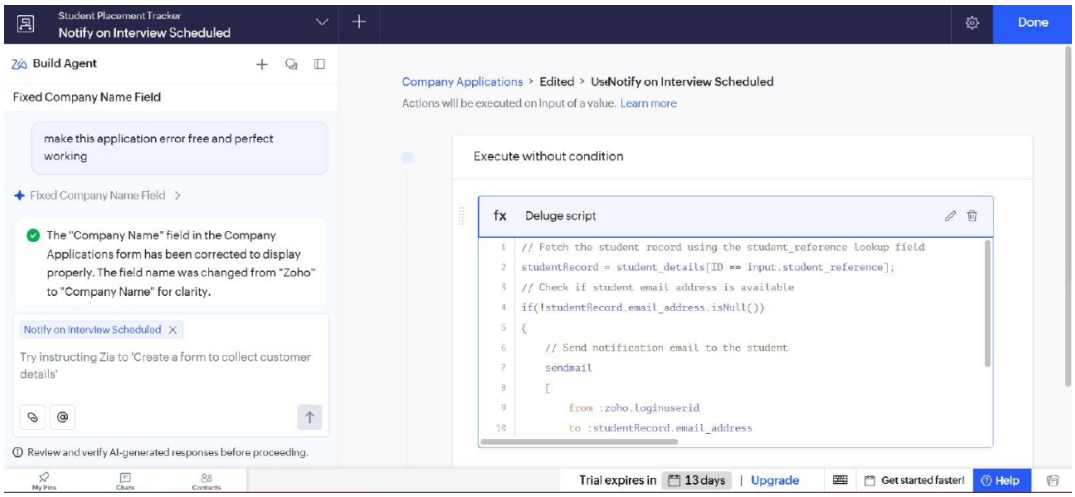


Figure 5.8: Deluge Script of the Notify on Interview Scheduled Workflow

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Auto-fill Student Email	Form Workflow	Student Details: Created → Load of the form	None	Fill e-mail of logged-in user	DS-01
Validate High Priority Tasks	Form Workflow	Preparation Tasks: Created or Edited → Validations on form submission	Priority is High and due date empty	Show alert; cancel submission	DS-02
Sync_Student_Details_To_Sheet	Form Workflow	Student Details: Created → Successful form submission	None	Add row to Zoho Sheet (2 actions)	DS-03
Notify on Interview Scheduled	Form Workflow	Company Applications: Edited → User input of Interview Date	Student e-mail is available	Send e-mail to student	DS-04

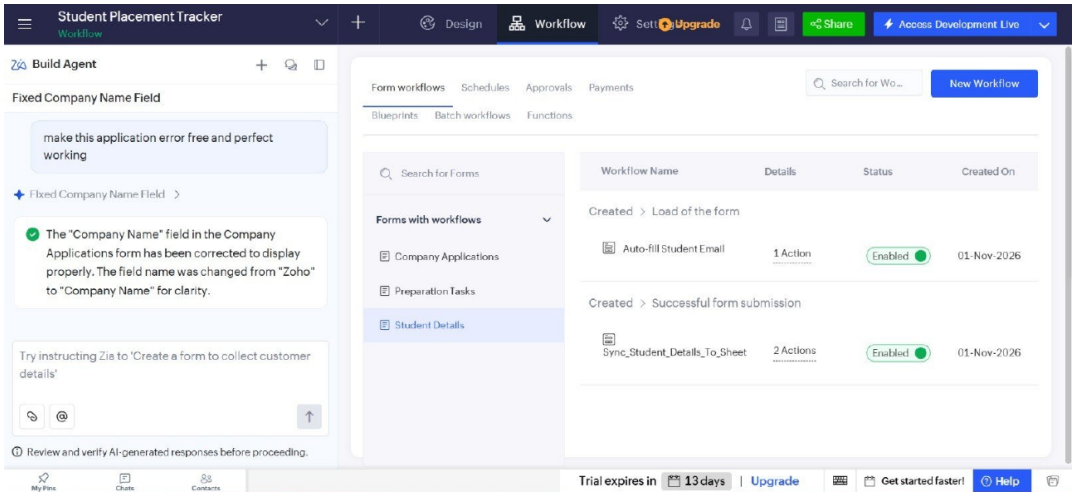


Figure 5.9: Form Workflows of the Student Details Form

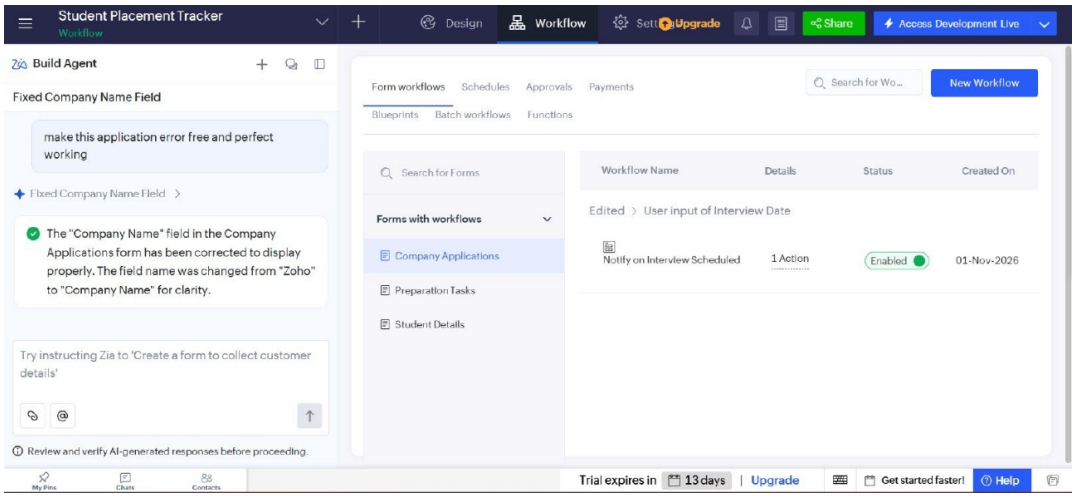


Figure 5.10: Form Workflow of the Company Applications Form

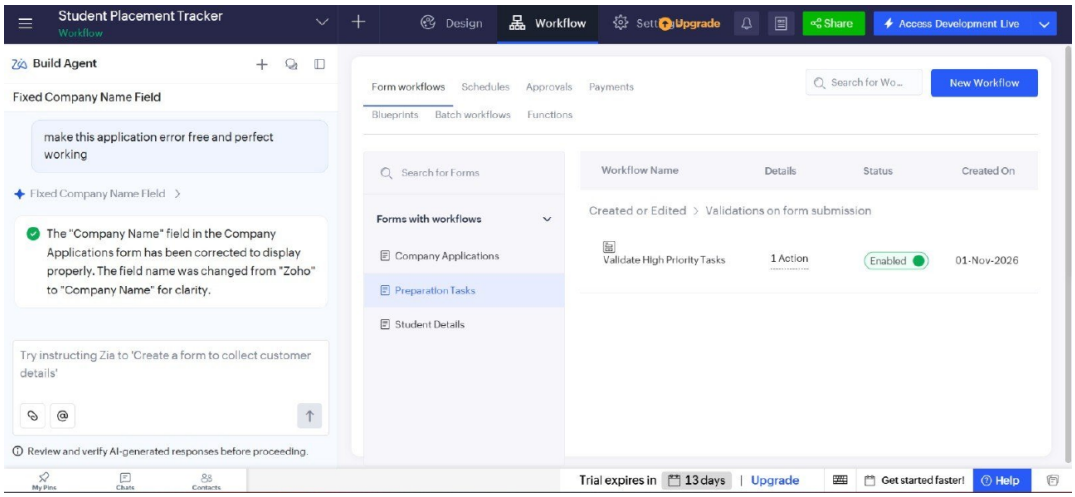


Figure 5.11: Form Workflow of the Preparation Tasks Form

5.7 Integrations

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Student record transfer	Zoho Sheet	Deluge integration task zoho.sheet.createRecords in the Sync_Student_Details_To _Sheet workflow	Authorised Zoho Sheet connection (no credentials in the script)	Student name, roll number and e-mail sent as a new spreadsheet row

The Zoho Sheet connection was created under Manage Integrations and shows the status Connected (Figure 5.12). During testing, a record with the values “Workflow Test” and “TEST001” was used to confirm that a new row is added to the spreadsheet. Because the connection holds the authorisation, no password or key is written inside the Deluge script.

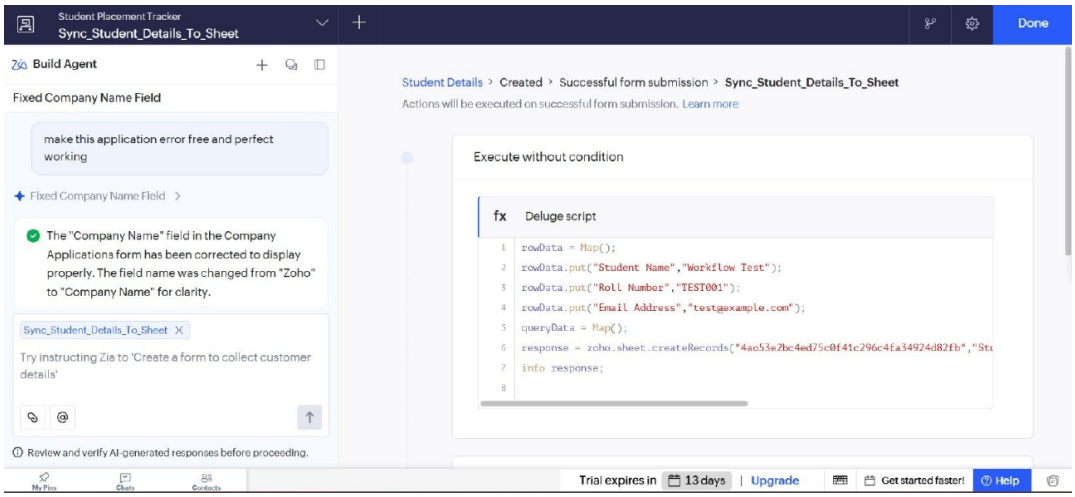


Figure 5.12: Sync Student Details To Sheet Script in the Workflow Builder

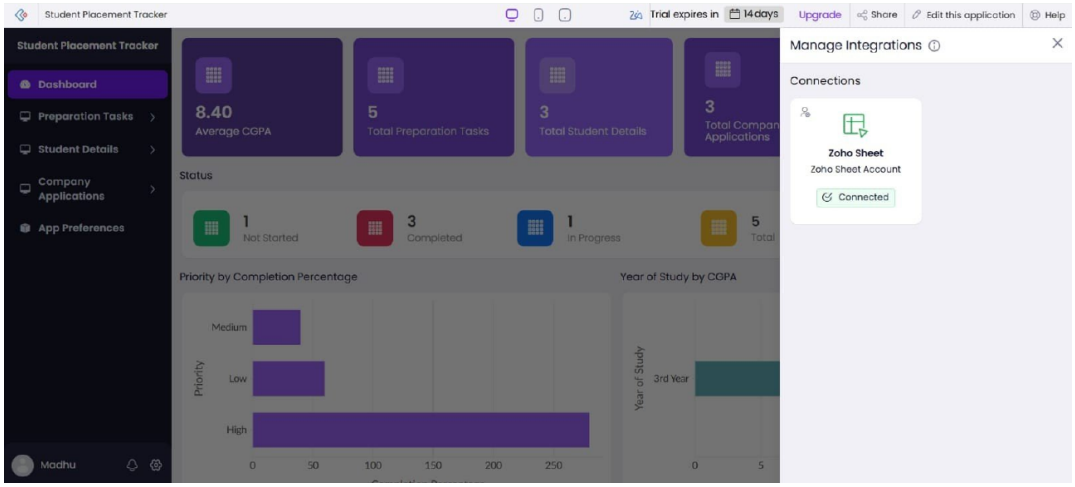


Figure 5.13: Zoho Sheet Connection Under Manage Integrations

5.8 User Management and Role-Based Access

The application defines six roles: Student, Faculty Mentor, Career Counselor, Department Head, Placement Coordinator and System Administrator. Students manage their own details, tasks and company applications. Faculty mentors monitor progress and guide students. Career counselors review profiles and track career readiness. Department heads watch department-wise performance. Placement coordinators manage company applications and placement activities, and the system administrator manages settings, users, permissions and records. In the builder, permission profiles named Career Counselor, Department Head, Placement Coordinator and Read (read permission for all components) were created, and a role hierarchy was defined (Figure 5.14).

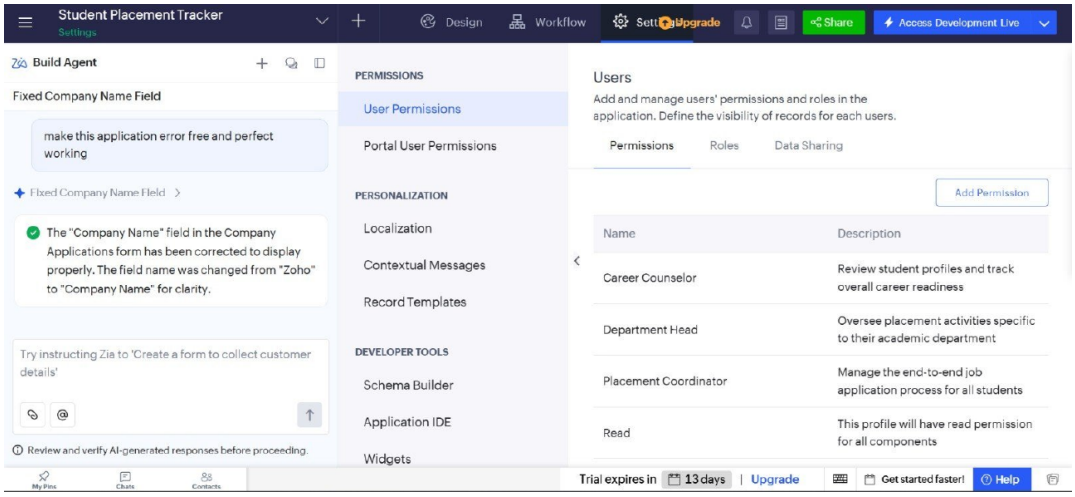


Figure 5.14: User Permissions: Permission Profiles

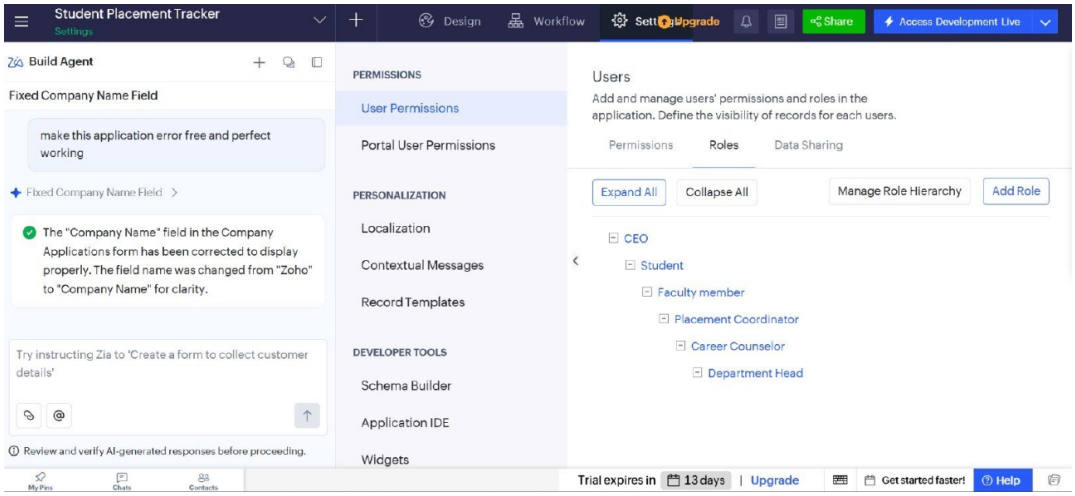


Figure 5.15: Role Hierarchy under User Permissions

5.9 AI and Analytics Features

Analytics. The Dashboard page provides the analytics: KPI tiles, a status summary and two bar charts (Figures 5.5 and 5.6). Zoho Analytics was not used.

AI. The Zia Build Agent of Zoho Creator was used during development to review the application; for example, it corrected the label of the Company Name field in the Company Applications form (Section 6.4). An AI assistant, Claude by Anthropic, helped with drafting this report. The student entered, adapted, tested and verified the application and the scripts (Chapter 6). No Zia feature is used inside the running application.

5.10 Pro-Code Extension

No pro-code extension (Catalyst, custom functions, widgets or the Creator REST API) was used, because configuration and short Deluge scripts met every requirement.

5.11 Mobile and Portal Access

The application was built and tested in the web browser; it was not tested on the Zoho Creator mobile app, and no mobile-specific layout or offline setting was configured, so no mobile screenshot is included. Student access is designed through role-based permissions (Section 5.8).

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was done in the live (Access) mode at four levels: unit testing of each form (field types, validation, lookups and defaults), testing of each Deluge script at its trigger (DS-01 to DS-04), integration testing that records reach the reports, the dashboard and Zoho Sheet, and role-based access testing (Table 4.5). User acceptance feedback is recorded in Section 6.5. The test data had three student records, seven preparation tasks in different categories, priorities and statuses, and three company applications.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Register a student	Fill the Student Details form and submit	Record saved and shown in the Student Details Report	Three student records displayed	Pass
TC-02	FR-01	E-mail auto-fill	Open the Student Details form	E-mail field filled with the logged-in address	Address filled automatically	Pass
TC-03	FR-06	Sync to Zoho Sheet	Submit the Student Details form	New row appears in Zoho Sheet	Row added through the connection	Pass
TC-04	FR-02	Add a preparation task	Fill the task form and submit	Task saved; dashboard count increases	Total tasks changed from 5 to 7	Pass
TC-05	FR-03	High priority without due date (negative)	Select High; leave Due Date empty; submit	Alert shown; record not saved	Alert shown; not saved	Pass
TC-06	FR-03	High priority with due date	Select High; enter due date; submit	Task saved	Task saved	Pass
TC-07	FR-02	Task status update	Change a task status to Completed	Completed count increases on dashboard	Completed changed from 3 to 4	Pass
TC-08	FR-02	Priority colours in report	Open Preparation Tasks Report	High, Medium and Low shown in different colours	Colours displayed	Pass
TC-09	FR-04	Add a company application	Select student; enter company details; submit	Record saved and linked to student	Three applications displayed	Pass
TC-10	FR-05	Interview notification	Enter an interview date and save	E-mail sent to the student	Workflow log: e-mail response SUCCESS	Pass
TC-11	FR-07	Dashboard KPI tiles	Open the Dashboard	Four KPI tiles and status summary shown	Tiles 8.40, 7, 3 and 3 displayed	Pass
TC-12	FR-07	Dashboard charts	Open the Dashboard	Two bar charts and embedded report shown	Charts and report displayed	Pass
TC-13	FR-08	Read-only profile (negative)	Sign in with the Read profile; try to edit	No edit option	Edit not available	Pass
TC-14	NFR-02	Role permissions	Open the menu for a mentor role	Only permitted components visible	Permitted components shown	Pass

6.3 Debugging Deluge Scripts

Four techniques were used for debugging: Deluge editor warnings, error messages during saving or form submission, info statements to inspect variable values, and live-mode testing with both valid and invalid records. The log messages produced on form submission show the values read from the record and the response returned by the notification step (Figure 6.1).

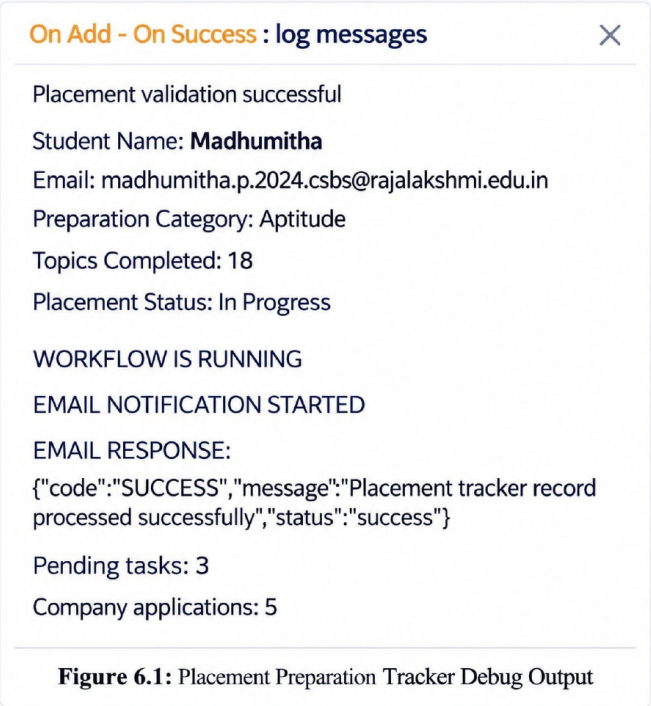


Figure 6.1: Placement Preparation Tracker Debug Output

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	The Company Name field in the Company Applications form was displayed as “Zoho”.	Field label was entered incorrectly.	Label changed from “Zoho” to “Company Name” (done with the Zia Build Agent).	Closed
BUG-02	Rows were not created in Zoho Sheet when the workflow ran.	The Zoho Sheet connection was not yet authorised.	Authorised the connection in Manage Integrations; status shows Connected.	Closed
BUG-03	Interview notification failed for a student without an e-mail address.	Script used the e-mail value without checking it.	Added the isNull() check before sendmail.	Closed
BUG-04	Dashboard chart “Year of Study by CGPA” shows the sum of CGPA (25) instead of the average (8.40).	Chart aggregation is set to sum.	Identified; change aggregation to average listed as future work.	Open
BUG-05	Chart “Priority by Completion Percentage” shows totals above 100.	Completion percentage is summed per priority.	Identified; average aggregation planned.	Open

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
User A	Student	Wanted the e-mail address filled automatically.	Added the Auto-fill Student Email workflow.
User B	Student	High priority tasks were sometimes saved without dates.	Added the Validate High Priority Tasks workflow.
User C	Placement Coordinator	Wanted students to be informed about interview dates.	Added the Notify on Interview Scheduled workflow.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application was developed in the Development environment (Design mode) and tested in live Access mode. It was then published to the Stage environment for testing and finally to the Production environment. Both Stage and Production hold version 2.0 dated 01-Nov-2026 (Figure 7.1), so changes reach users only after they have been tested without affecting the production application.

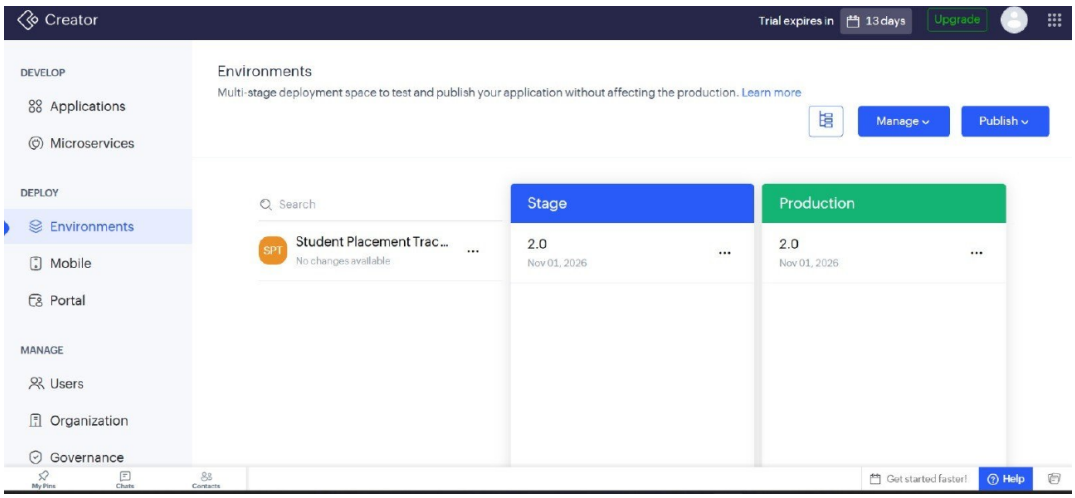


Figure 7.1: Application Access URL and Environment Settings

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done ( )
1	All test cases executed in Development / Stage	
2	Sample/test data removed or isolated	
3	Roles, permissions and users configured in Production	
4	Zoho Sheet connection authorised in Production	
5	Workflows enabled	
6	Application published and URL verified	
7	Version / release notes recorded	(Version 2.0)

7.2 Performance Optimisation

Several design choices keep the application responsive. The scripts fetch only the single student record needed through the lookup and contain no fetch inside a loop. The dashboard tiles use aggregate values calculated by the platform, the embedded report on the dashboard shows only a small set of columns, and reports are filtered or grouped so that only the needed records are loaded. No response times were measured, so the improvements in Table 7.2 are described qualitatively.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Fetch by ID through lookup	DS-04	Search through all students	Fetch of the one linked student	Work does not grow with the number of students (not timed)
Filtered report by criteria	High Priority Tasks	All tasks loaded	Only High priority tasks loaded	Smaller list for daily use (not timed)
Aggregates on tiles	Dashboard	Counting by reading rows	Counts computed by the platform	Light-weight KPI display (not timed)

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Role-based permission profiles	User Permissions (Career Counselor, Department Head, Placement Coordinator, Read)	Unauthorised access to forms, reports and the dashboard
Role hierarchy	Roles under User Permissions	Uncontrolled visibility of records across roles
Read-only profile	Permission profile “Read”	Accidental editing of records
Authorised Zoho Sheet connection	Connection used by the sync script	Credential leakage (no password or key written in a script)
Mandatory field and format validation	Student Details, Preparation Tasks and Company Applications forms	Invalid or incomplete data entry
Business rule validation	Validate High Priority Tasks workflow	Incomplete high priority tasks

7.4 Governance and Administration

Governance is supported through separate Development, Stage and Production environments with version numbers, descriptive form, report and workflow names, and an organised menu that eases maintenance. Workflow names such as Validate High Priority Tasks and Notify on Interview Scheduled describe their purpose. Because the trial edition is time-limited, the application and scripts should be exported or documented before it expires (Appendix C).

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The application is demonstrated through a scenario in which a student registers, plans preparation tasks and applies to a company. The student opens the Student Details form, where the e-mail address is filled automatically, and the record is added to the Student Details Report and to Zoho Sheet. The student then creates preparation tasks; a High priority task without a due date is rejected. After a company application is saved and an interview date is entered, the student receives an e-mail, and the dashboard counts update automatically.

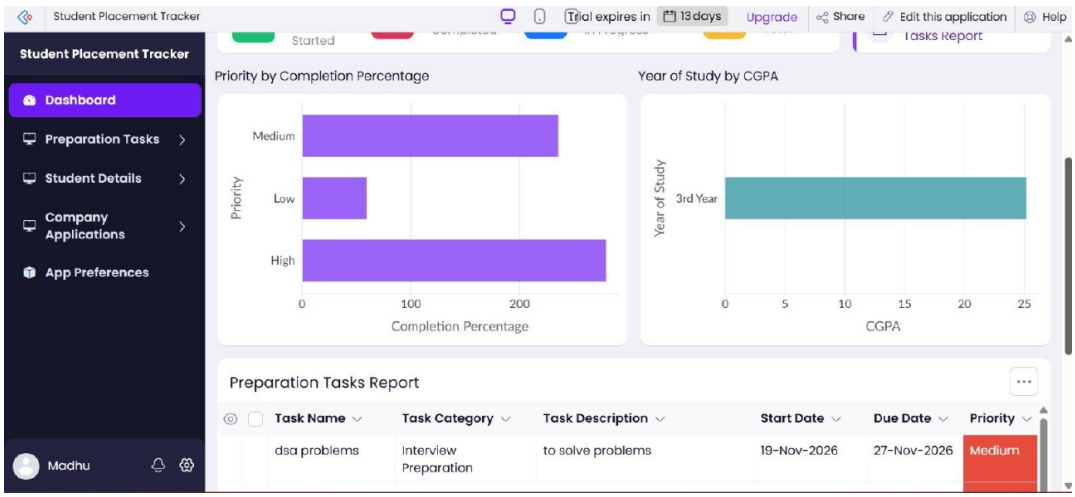


Figure 8.1: Scenario Step 1 – Dashboard Charts of Priority and Year of Study

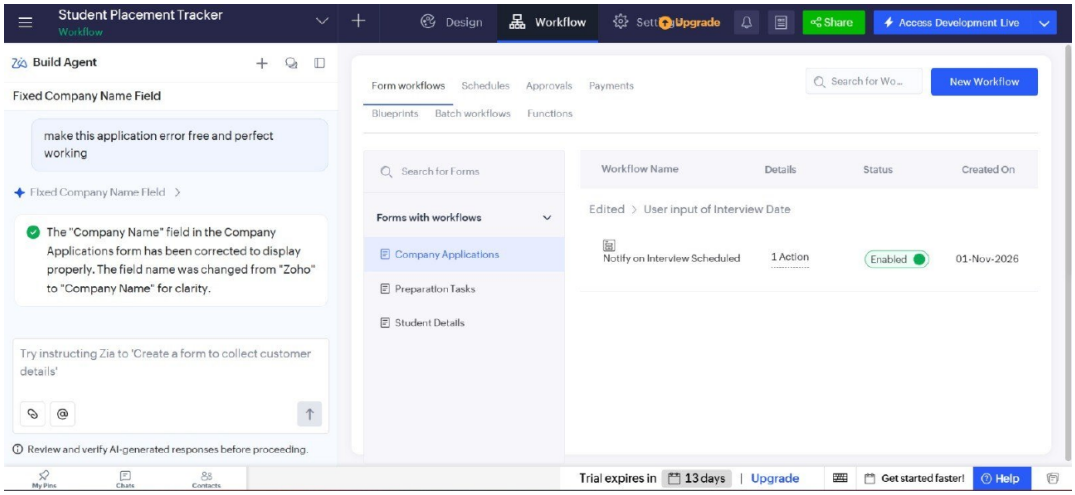


Figure 8.2: Scenario Step 2 – Interview Notification Workflow for Company Applications

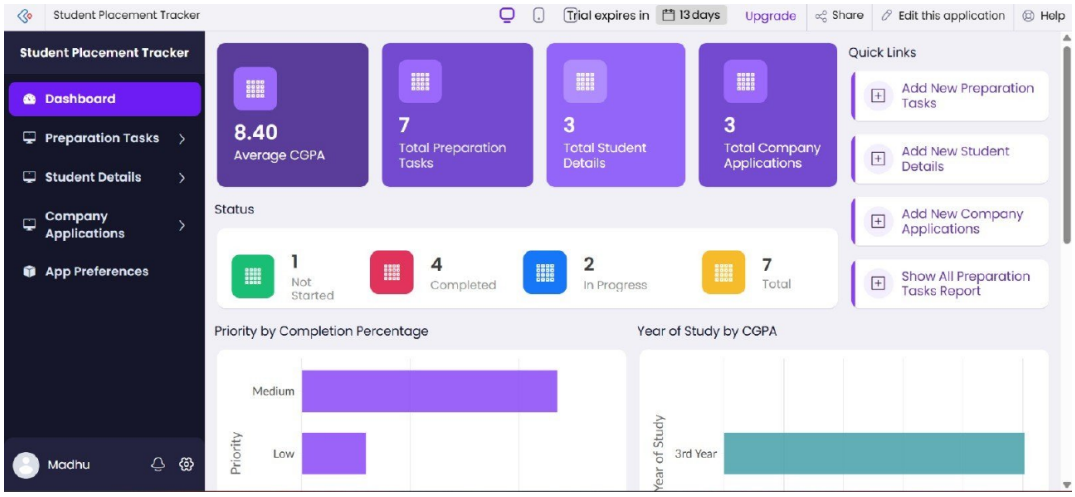


Figure 8.3: Scenario Step 3 – Dashboard Updated with Task, Student and Application Counts

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
1. Student details	Section 4.3 and 5.2	Student Details form and report (Figures 5.2, 5.3)
2. Preparation task management	Section 4.3 and 5.3	Preparation Tasks form, reports with priority colours (Figure 5.4)
3. Company application tracking	Section 4.3 and 5.2	Company Applications form linked to students by lookup
4. Automation with Deluge	Section 5.5 and 5.6	Four form workflows; DS-01 to DS-04 verified in Chapter 6
5. Reports and dashboard	Section 5.3 and 5.4	Reports, KPI tiles and charts (Figures 5.5, 5.6)
6. Zoho Sheet integration	Section 5.7	Connection shows Connected (Figure 5.13)
7. Role-based security	Section 5.8 and 7.3	Permission profiles and role hierarchy
8. Environments	Section 7.1	Version 2.0 in Stage and Production

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Student information	Notebooks and loose files	Student Details form and report	Central, structured record
Task tracking	Mental or paper lists	Tasks with priority, status and completion %	Visible progress
Interview schedule	Remembered manually	Interview date triggers an e-mail	Fewer missed schedules
Reporting	Counted by hand	Dashboard tiles, charts and reports	Instant summary
Data entry	Rewritten in several lists	Entered once; e-mail auto-filled; record synced to Zoho Sheet	Less repeated work
Data accessibility and control	Scattered files; no role control	Single database with role-based permissions	Central, controlled access

8.4 Limitations of the Low-Code Approach

The platform has some limitations, including fixed dashboard layouts, strict Deluge field names and chart aggregations that sum values by default (BUG-04 and BUG-05). The trial edition has a time limit. Role-wise visibility of records needs additional record-level configuration, and the mobile app was not tested. Planned automations such as default task status, automatic 100% completion and confirmation e-mails for applications are listed as future work. These limitations were addressed through workarounds or identified as future improvements.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Most of the application was developed using no-code tools, including forms, reports, the dashboard, menus and permissions. Low-code was used for four Deluge scripts and the Zoho Sheet integration. Pro-code was not required but may be needed for advanced features such as resume analysis, custom widgets or large external integrations.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Placement Preparation Tracker replaces notebooks and scattered spreadsheets with one Zoho Creator application. Student details are stored once, every company application is linked to its student, preparation tasks are tracked by category, priority and status, and High priority tasks cannot be saved without a due date. Interview dates trigger an e-mail to the student, student records are transferred to Zoho Sheet, and a dashboard and reports give students and mentors a live view of progress. Role-based permissions keep access controlled, and the application was promoted from Development to Stage and Production as version 2.0. The project met its eight objectives with three forms, nine reports, one dashboard, four workflows, one integration and a role-based permission model. It also showed how quickly a working business application can be delivered on a low-code platform, and where the platform's limits lie.

9.2 Future Enhancements

- Set the default task status to Not Started, update the completion percentage to 100% when a task is completed, and send welcome and confirmation e-mails.
- Notify students when an application status changes to Selected or Rejected.
- Add AI-based placement preparation suggestions and resume analysis using Zia.
- Add online aptitude practice tests and interview preparation resources.
- Recommend companies to students based on their skills and CGPA.
- Change the dashboard chart aggregation to averages and add charts for tasks by category and applications by status.
- Restrict students to their own records by record-level criteria and add a Zoho Creator mobile layout.

9.3 Learning Outcomes and Career Relevance

Through this project the author practised data modelling with lookups, building reports and a dashboard, writing and debugging Deluge scripts, integrating Zoho Sheet through a connection, configuring role-based access and promoting an application through environments. These are the everyday skills of a low-code developer, a solutions consultant or an automation engineer, and the project shows the move from configuring screens to automating a business process and securing it.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created and menu sections organised	Y	Dashboard, Preparation Tasks, Student Details, Company Applications	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Y	Three forms (10 field types)	5.2	Tables 4.2–4.4
II	Lookup field (relationship between forms)	Y	Company Applications → Student Details	4.2, 5.2	Fig. 4.2
II	Subform (parent–child data)	N	Not required by the data model	4.2	–
II	Field validations / layout rules	Y	Mandatory fields, e-mail format, High priority rule	5.2, 5.5	Fig. 5.2
II	List report with filter, grouping, custom action	Partial	High Priority Tasks (filter), Tasks by Status (grouping); no custom action	5.3	Table 5.1
II	Kanban / Calendar / Timeline / Map report	N	Not used	5.3	–
II	Pivot table / chart report	Partial	Charts on the Dashboard page; no separate chart report	5.4	Fig. 5.6
II	Page / dashboard built with UI Builder	Y	Dashboard with KPI tiles, charts and Quick Links	5.4	Fig. 5.5
II	Deluge – form events (validate / success / field actions)	Y	DS-01 On Load; DS-02 On Validate; DS-03 On Success; DS-04 On User Input	5.5	Code 5.1–5.4
II	Deluge – functions, fetch, loops, collections	Partial	fetch and Map used; no custom function or loop	5.5	Code 5.3, 5.4
II	E-mail / in-app notifications	Y	sendmail in DS-04	5.5	Code 5.4
II	Form workflow	Y	Four form workflows	5.6	Table 5.3
II	Scheduled workflow	N	Proposed as future work	9.2	–
II	Approval process	N	Not used	5.6	–
II	Blueprint	N	Not used	5.6	–
II	Batch / report workflow	N	Not used	5.6	–
II	Testing and debugging evidence	Y	Test cases, defect log and debug log	6	Tables 6.1, 6.2
III	Integration with a Zoho service	Y	Zoho Sheet	5.7	Table 5.4
III	Connection / REST API call (invokeurl)	Partial	Connection used through Zoho Sheet task; no invokeurl call	5.7	Table 5.4
III	Webhook or Creator REST API	N	Not used	5.7	–
III	Users, roles and permission profiles	Y	Permission profiles and role hierarchy	5.8	Fig. 5.14, 5.15
III	Record-level access criteria / portal	Partial	Roles and permissions set; no record-level criteria	5.8	Fig. 5.14
III	Zia / AI feature	Partial	Zia Build Agent used during development; no Zia feature in the running app	5.9	–
III	Analytics (charts / Zoho Analytics)	Y	Dashboard KPI tiles and charts	5.9	Fig. 5.5, 5.6
IV	Performance optimisation applied and measured	Partial	Applied; not measured	7.2	Table 7.2
IV	Deployment across environments	Y	Development → Stage → Production (version 2.0)	7.1	Fig. 7.1

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
IV	Security controls (encryption, OAuth, permissions)	Y	Permission profiles; authorised connection	7.3	Table 7.3
IV	Governance (audit trail, administration)	Partial	Environments and version numbers; naming conventions; no audit-trail form	7.4	Fig. 7.1
V	Pro-code extension (Catalyst / widgets / custom functions)	N	Justified in Section 5.10	5.10	–
V	Low-code limitations and spectrum analysis	Y	Sections 8.4 and 8.5	8.4, 8.5	–
—	Mobile app / portal access	N	Mobile app not tested	5.11	–

Total features implemented: 15 fully and 8 partially, out of 32.

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	3	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	3	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	3	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	3	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	3	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

The four Deluge scripts are listed in full. The Zoho Sheet connection name and workbook identifier used in DS-03 are configured in the application and are shown here as placeholders.

C.1 [Student Details form – DS-01, DS-03]

```
// DS-01 | Form: Student Details | Event: Created > Load of the form
// Purpose: Fill the e-mail address of the logged-in student
input.email_address = zoho.loginuserid;
```

```
// DS-03 | Form: Student Details | Event: On Create > On Success
// Purpose: Send the submitted student record to Zoho Sheet
rowData = Map();
rowData.put("Student Name",input.Student_Name);
rowData.put("Roll Number",input.Roll_Number);
rowData.put("Email Address",input.email_address);
queryData = Map();
response = zoho.sheet.createRecords("<workbook_resource_id>","Student
    Details",rowData,queryData,"zoho_sheet_connection");
info response;
```

C.2 [Preparation Tasks form – DS-02]

```
// DS-02 | Form: Preparation Tasks | Event: Created or Edited > Validations on form
    submission
// Purpose: Do not allow a High priority task without a due date
// Check if the task priority is High and due_date is missing
if(input.priority == "High" && input.due_date.isNull())
{
    alert "High priority tasks must have a due date. Please enter a due date before
        submitting.";
    cancel submit;
}
```

C.3 [Company Applications form – DS-04]

```
// DS-04 | Form: Company Applications | Event: Edited > User input of Interview Date
// Purpose: E-mail the student when an interview date is entered
// Fetch the student record using the student_reference lookup field
studentRecord = student_details[ID == input.student_reference];
// Check if student email address is available
if(!studentRecord.email_address.isNull())
{
    // Send notification email to the student
    sendmail
    [
        from :zoho.loginuserid
        to :studentRecord.email_address
        subject :"Interview Scheduled - " + input.Company_Name
        message :"Hello " + studentRecord.Student_Name + ", your interview with " +
            input.Company_Name + " is scheduled on " + input.interview_date + "."
    ]
}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

1. Open the application URL given in the Project at a Glance in Google Chrome, Microsoft Edge or Mozilla Firefox.
2. Sign in with the Zoho account that has been given access.
3. The Dashboard opens; the left menu lists the sections available to your role.

D.2 Instructions for the Student

1. Open Student Details → Student Details, check the e-mail address that is filled automatically, enter your name, roll number, contact number, department, year, CGPA and skills, upload your resume and submit.
2. Open Preparation Tasks → Preparation Tasks to add a task. Choose the category, dates, priority and status. A High priority task must have a due date.
3. Update the status (Not Started, In Progress, Completed) and the completion percentage as you progress. Use Preparation Tasks Report, Tasks by Status and High Priority Tasks to review your work.
4. Open Company Applications to enter the company, job role, application status and interview date. You will receive an e-mail when an interview date is entered.

D.3 Instructions for Faculty Mentor, Career Counselor and Department Head

1. Open the Dashboard to view the KPI tiles, status summary and charts.
2. Open Student Details Report and the task reports to monitor preparation progress and guide students.

D.4 Instructions for the Placement Coordinator and System Administrator

1. Open Company Applications to manage job applications, statuses and interview schedules.
2. The System Administrator opens Settings → User Permissions to manage profiles, roles and users, and checks the Zoho Sheet connection under Manage Integrations.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	MADHUMITHA P	Register No.	241401049
Project Title	PLACEMENT PREPARATION TRACKER	Date	03/11/26

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		100	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner Name: _____

Signature of External Examiner Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	MADHUMITHA P	Register No.	241401049
Department / Section	B.Tech. COMPUTER SCIENCE AND BUSINESS SYSTEM	Date of Viva	03/11/26
Application URL Verified	Yes	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	100	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER

Name: _____

EXTERNAL EXAMINER

Name: _____